

CSC-1700: Analysis of Algorithms

Fall 2025

Instructor: Nadia Niknami
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Office: Mendel 292 D, Dept. of Computing Sciences
Office hours: Fridays 1:00–3:00 PM (and by appt.)
Class Meeting: Section 003: MWF 10:40 am- 11:30 am
MW in Mendel 259 and F in G88

Section 004: MWF 11:45 am- 12:35 am
MW in Mendel 259 and F in 141

Course Description

This course introduces the theory underlying computer algorithms. Students successfully completing this class will have a basic understanding of how algorithms for problem solving on computers are designed and analyzed and will be prepared to begin independently designing algorithms to solve new problems. The course begins with a tutorial of the basic algorithmic analysis techniques, continues with a study of basic algorithms for solving common computing problems, and concludes with a look at computational complexity and approximate solutions.

Topics include efficiency classifications and mathematical analysis of recursive and non-recursive algorithms; design techniques: brute force, greedy, decrease-and-conquer, divide-and-conquer, transform-and-conquer, dynamic programming, iterative improvement, backtracking, branch-and-bound, space and time tradeoffs; NP-completeness; approximation algorithms; computational problems: sorting, searching, string processing, graphs and arithmetic.

Course Objectives

- Establish an understanding of fundamental techniques for the design and analysis of algorithms.
- Establish an understanding of efficiency classifications and mathematical analysis of recursive and nonrecursive algorithms.
- Apply design and analysis techniques to important problems from various areas of computing including sorting, searching, string processing, graphs, arithmetic, and matrix algebra.

Prerequisites

- CSC 1300 (or MAT 2600)
- CSC 1052 (or ECE 2620)
- If you lack the prerequisite, please contact the instructor

Course Resources

- (Required) Introduction to the Design and Analysis of Algorithms, 3rd edition, by Anany Levitin, ISBN-10: 0-13-231681-1

- (Recommended) Algorithmic Puzzles by Anany and Maria Levitin, ISBN-10: 0-19-974044-5
- Blackboard

Course Requirements

- **Assignments:** Weekly assignments include readings and exercises (with hints and solutions provided). While the assignments will not be graded, they provide critical practice for the course material and help students perform well on quizzes and exams.
- **Quizzes:** There will be five 45-minute quizzes that include questions from lecture notes and assignments. They will be administered at the beginning of the class, so make sure to show up on time to class. Missed quizzes cannot be made up without penalty unless you have a legitimate, documented reason for missing the quiz and you notify the instructor in advance (as the situation permits). Missed quizzes without documentation result in a 12 point deduction.
- **Coding Contests:** There will be five coding contests administered through hacerrank.com. These coding problems are meant to reinforce the algorithms learned in class and encourage algorithmic thinking when coding. There is a 3 point late penalty per business day up to 21 points. After that you can submit at any time for a max score of 79. Late points do not accumulate over the weekend.
- **Tests:** one midterm and one final exam. Tests will be closed books, closed notes, closed electronic devices. However, you are allowed to bring one sheet of paper (letter size) with any handwritten information you think will help you during the exam. Such notes may not be shared during the exam.

Grading Scheme

Component	Weight
Quizzes	25%
Code Contest	25%
Midterm Exam	25%
Final Exam	25%

On a 100-point scale, you can expect the following letter grades: A (94–100), A– (90–93), B+ (88–89), B (84–87), B– (80–83), C+ (78–79), C (74–77), C– (70–73), D+ (60–69), D (54–59), D– (50–53), F (<50).

Policies

1. **Attendance and Class Participation:** Class discussions may include examples and information not covered in the text, so attendance is highly recommended. Each student is responsible for all announcements and material covered during any class missed. In addition to the attendance policy stated above, students are entitled to two excused absences for any reason that may contribute to their personal wellness. Students must advise the instructor by email before class of their intent to utilize a Personal Day as the reason for their absence.

A Personal Day will not be approved retroactively. Students may, but are not required, to provide additional information regarding their absence. Additionally, Personal Days may not:

- be used on consecutive class days;
- be used in the same week;
- be used immediately preceding or following a University holiday or break period;
- be used on days when exams, presentations or other major assignments are scheduled.

A Personal Day does not grant an automatic extension for items due. Students remain responsible for all assignments, exams, presentations, etc., due on that date. It is in the instructor's discretion to determine whether any extension is appropriate given individual circumstances. For more information, see [Class Attendance Policy](#).

2. **Quiz Makeup Policy:** Missed quizzes cannot be made up without penalty unless you have a legitimate, documented reason for missing the quiz and you notify the instructor in advance (as the situation permits). Missed quizzes without documentation result in a 12 point deduction.
3. **Coding Contests:** There is a 3 point deduction for each business day a contest is late up to 21 points. After that you can submit at any time for a maximum score of 79.
4. **No extra credit:** In fairness to all students, I do not offer extra credit assignments to individuals. If extra credit is provided, it will be provided to the entire class and will be announced during class and not on Blackboard. Please speak with me if you are having problems. I will be happy to help you.
5. **Use of Artificial Intelligence (AI):** Artificial intelligence tools (such as ChatGPT, Copilot, Gemini, etc.) may be helpful for brainstorming, clarifying concepts, or improving writing clarity. However, their use in this course is permitted only under the following conditions:
 - If you use an AI tool in preparing any part of your work, you must clearly indicate what tool you used and how.
 - You may use AI tools only for idea generation, outlining, or editing suggestions, not for producing final answers, code, or solutions that you submit for grading.
 - AI tools are strictly prohibited. Use of AI during any in-class or take-home exam will be considered academic dishonesty.
6. **Office for Disability Services (ADS) and Learning Support Services (LSS):** It is the policy of Villanova to make reasonable academic accommodations for qualified individuals with disabilities. All students who need accommodations should go to Clockwork for Students via myNOVA to complete the Online Intake or to send accommodation letters to professors. Go to the LSS website learningsupportservices.villanova.edu or the ADS website [ADS](#) for registration guidelines and instructions. If you have any questions, contact LSS at 610-519-5176 or learning.support.services@villanova.edu, or ADS at 610-519-3209 or ods@villanova.edu.
7. **Academic Integrity:** All students are expected to uphold Villanova's Academic Integrity Policy and Code. Any incident of academic dishonesty will be reported to the Dean of the College of Liberal Arts and Sciences for disciplinary action. See the University's policy: [Academic Integrity Policy and Code](#).
If a student is found responsible for an academic integrity violation that results in a grade

penalty, they may not WX the course unless approved to WX for significant medical reasons. Documentation is required and requests will be considered.

8. **Absences for Religious Holidays:** Villanova University makes every reasonable effort to allow members of the community to observe their religious holidays, consistent with the University's obligations, responsibilities, and policies. Students who expect to miss a class or assignment due to the observance of a religious holiday should discuss the matter with their professors as soon as possible, normally at least two weeks in advance. Absence from classes or examinations for religious reasons does not relieve students from responsibility for any part of the course work required during the absence. Policy: [Religious Holidays](#).

I am always available and always approachable! I encourage you to email (nniknami@villanova.edu) or come see me if you have a problem or question.